

Haoming Li

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RESEARCH INTERESTS

I work on **computer-use agents (CUAs)** for long-horizon tasks and **persona-aware CUAs**, in collaboration with MatraIX. Currently, I'm interested in **persona AI** and **agentic memory**, which I expect persona agents to depend on. Earlier, I worked on scientific summarization and LLM planning.

EDUCATION

The University of Texas at Dallas , Ph.D. in Computer Science, GPA 3.83/4.00	Aug 2025 – Present
Emory University , M.S. in Computer Science, GPA 3.97/4.00	Aug 2022 – May 2024
Renmin University of China , B.S. in Data Science and B.Econ. in Finance	Sep 2018 – Jun 2022

PUBLICATIONS

[†] equal contribution

- [1] Xu Hu[†], **Haoming Li**[†], Yuxiang Wang, Jinxin Yang, Kangshuo Li, Licheng Wang, Dingyi Kang, Jiayu Song, Tianhao Wang, Zhanhua Hu, Xu Yang, Qiannan Li, Bingzhe Li, Feng Chen. *CUA-Recovery: Benchmarking Error Recovery in Long-Horizon Computer-Use Agents*. Under review at **ICLR 2027**.
- [2] Xu Hu, **Haoming Li**, Sabbir Ahmed, Md Nahiyen Uddin, Qiannan Li, Jessica Ouyang, Latifur Khan, Feng Chen, Bingzhe Li. *Toward Trustworthy Computer-Use Agents: Risk Propagation, Evaluation Gaps, and Human Governance*. Under review at **ACM Computing Surveys**.
- [3] Weili Shi, **Haoming Li**, Sheng Li. *Enhancing Dynamic Text-Attributed Graph Modeling via Instruction Tuning*. **Findings of AACL-IJCNLP 2026** (to appear).
- [4] **Haoming Li**, Jessica Ouyang. *GRASP: Graph-Reasoning Aided Survey Planning for High-Fidelity Related Work Generation*. **Findings of ACL 2026**. [paper]
- [5] **Haoming Li**, Zhaoliang Chen, Songyuan Liu, Yiming Lu, Fei Liu. *Systematic Analysis of LLM Contributions to Planning: Solver, Verifier, Heuristic*. arXiv:2412.09666, 2024.
- [6] **Haoming Li**, Zhaoliang Chen, Jonathan Zhang, Fei Liu. *LASP: Surveying the State-of-the-Art in Large Language Model-Assisted AI Planning*. arXiv:2409.01806, 2024.
- [7] Yikun Wang, Rui Zheng, **Haoming Li**, Qi Zhang, Tao Gui, Fei Liu. *RESCUE: Ranking LLM Responses with Partial Ordering to Improve Response Generation*. **ACL 2024 Student Research Workshop**. [paper]

RESEARCH EXPERIENCE

The University of Texas at Dallas , Research Assistant (advisors: Feng Chen, Bingzhe Li)	Jun 2025 – Present
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- **CUA-Recovery** (co-first author) [1]: designed *ReRail*, the task and trajectory synthesis framework behind the benchmark (2,250 human-verified erroneous states for testing, 8.6K trajectories for training).
- Analyzed six agents at six takeover depths (best error-awareness rate: 44.0%) and fine-tuned Qwen3.5-35B-A3B on the training split with SFT, raising its error-awareness rate from 19.9% to 35.2% on held-out states.
- **Trustworthy CUA survey** (second author) [2]: reviewed and organized the literature on CUA security, privacy, reliability, controllability, and accountability.
- **GRASP** (first author, advised by Jessica Ouyang) [4]: designed, implemented, and evaluated a graph-reasoning planner for related-work generation.

University of Virginia, Reasoning & Knowledge Discovery Lab , Research Assistant (advisor: Sheng Li)	Jun 2024 – May 2025
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- Instruction-tuned LLMs to encode and reason over dynamic text-attributed graphs [3].

Emory University, NLP Group , Research Assistant (advisor: Fei Liu)	Mar 2023 – May 2025
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- **LLMs for planning** (first author) [5, 6]: led an empirical study of LLMs as solver, verifier, and heuristic function in planning, and a survey of LLM-assisted planning.
- **RESCUE** [7]: trained a response ranker from partial orderings over candidate LLM outputs to improve response generation.

SERVICE AND SKILLS

Reviewer: NeurIPS 2026. **Languages:** English (fluent), Chinese (native).

Programming: Python, PyTorch, C/C++, SQL, Bash. **Tools:** Hugging Face Transformers, vLLM, Docker, Spark.